A graduate student is conducting an experiment to understand how learning occurs in a feedforward neural network using backpropagation. The objective of the experiment is to manually compute one complete training iteration, including the forward pass, error calculation, and backward pass using the chain rule.

The neural network designed for this task consists of two input neurons, three neurons in a single hidden layer, and two output neurons. Each neuron in the hidden layer receives inputs from both input neurons, and each output neuron receives inputs from all three hidden neurons. The sigmoid activation function is used at both the hidden and output layers. To simplify the model, the student uses one shared bias value for all hidden neurons and one shared bias value for all output neurons.

For this experiment, the student selects a single training example. The input to the network is defined as $x_1 = 0.6$and $x_2 = 0.1$, representing two features of the data. The expected target output for this input is $t_1 = 1$and $t_2 = 0$, indicating that the input belongs strongly to the first class and not to the second. The learning rate for updating the network parameters is set to $\eta = 0.1$.

The initial weights between the input layer and the hidden layer are assigned as follows: the weights connecting $x_1$to the hidden neurons $h_1, h_2, h_3$are $w_{11} = 0.2$, $w_{12} = 0.4$, and $w_{13} = -0.5$, respectively. Similarly, the weights connecting $x_2$to the hidden neurons are $w_{21} = -0.3$, $w_{22} = 0.1$, and $w_{23} = 0.2$.

The weights between the hidden layer and the output layer are initialized such that the connections from hidden neurons $h_1, h_2, h_3$to output neuron $o_1$are $v_{11} = 0.3$, $v_{21} = -0.2$, and $v_{31} = 0.5$, while the connections to output neuron $o_2$are $v_{12} = -0.4$, $v_{22} = 0.2$, and $v_{32} = -0.3$.

The bias applied uniformly to all hidden neurons is $b_h = 0.25$, and the bias applied uniformly to both output neurons is $b_o = -0.15$.

The student now wants to verify the learning mechanism of the network by performing all calculations manually. You are required to carry out one complete iteration of training for this neural network. This includes computing the forward propagation by determining the net input and activated output at each neuron in the hidden and output layers, followed by calculating the total error using the mean squared error function.

After computing the error, you must perform the backward propagation step using the chain rule. This involves calculating the error terms (deltas) for both output neurons and then propagating these errors backward to compute the error terms for each hidden neuron. Using these error terms, update all the weights in the network, including both the weights between the hidden and output layers and the weights between the input and hidden layers. Finally, update the bias values for both the hidden layer and the output layer.

All intermediate steps, mathematical expressions, and numerical calculations must be clearly shown, and the use of the chain rule in backpropagation must be explicitly demonstrated. Only one full iteration of forward and backward propagation is required.